Let us consider the formation of a metal complex ML_6 in aqueous medium.(Charge on the metal ion is ignored) complex formation may occur in single step or step by step.

If ligands added to the metal ion in single step, then

$$[MS_6] + 6\,L \rightleftharpoons [ML_6] + 6\,S$$

$$\beta_{overall} = \frac{[ML_6]\,[S]^6}{[MS_6]\,[L]^6}$$

$\beta_{overall}$ is called as overall stability constant. As solvent is present in large excess, its concentration in the above equation can be ignored.

$$\therefore\ \beta_{overall} = \frac{[ML_6]}{[MS_6]\,[L]^6}$$

If these six ligands are added to the metal ion one by one, then the formation of complex $[ML_6]$ can be supposed to take place through six different steps as shown below. Generally step wise stability constants are represented by the symbol k.

$$[MS_6] + L \rightleftharpoons [MS_5L] + S \qquad k_1 = \frac{[MS_5L]}{[MS_6][L]}$$

$$[MS_5L] + L \rightleftharpoons [MS_4L_2] + S \qquad k_2 = \frac{[MS_4L_2]}{[MS_5L][L]}$$

$$[MS_4L_2] + L \rightleftharpoons [MS_3L_3] + S \qquad k_3 = \frac{[MS_3L_3]}{[MS_4L_2][L]}$$

$$[MS_3L_3] + L \rightleftharpoons [MS_2L_4] + S \qquad k_4 = \frac{[MS_2L_4]}{[MS_3L_3][L]}$$

$$[MS_2L_4] + L \rightleftharpoons [MSL_5] + S \qquad k_5 = \frac{[MSL_5]}{[MS_2L_4][L]}$$

$$[MSL_5] + L \rightleftharpoons [ML_6] + S \qquad k_6 = \frac{[ML_6]}{[MSL_5][L]}$$

In the above equilibrium, the values k_1, k_2, k_3, k_4, k_5 and k_6 are called step wise stability constants. By carrying out small a mathematical manipulation, we can show that overall stability constant β is the product of all step wise stability constants k_1, k_2, k_3, k_4, k_5 and k_6.

$$\beta = k_1 \times k_2 \times k_3 \times k_4 \times k_5 \times k_6$$

On taking logarithm both sides

$$\log(\beta) = \log(k_1) + \log(k_2) + \log(k_3) + \log(k_4) + \log(k_5) + \log(k_6)$$

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